Code No: 124CN **R15**

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B.Tech II Year II Semester Examinations, November/December - 2020
COMPUTER ORGANIZATION
(Computer Science and Engineering)

Time: 2 hours **Max. Marks: 75**

Answer any Five Questions
All Questions Carry Equal Marks
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1. An instruction is stored at location 300 with its address field at location 301. The address field has the value 400. A processor register R1 contain the number 200. Evaluate the effective address if the addressing mode of the instruction is
   a) Direct b) Immediate c) Relative [5+5+5]

2. Make a comparison between the isolated I/O and memory-mapped I/O. [15]

3. Describe Direct mapping technique of Cache memory. A digital computer has a memory unit of 64K ×16 and the cache memory of 1K word. The cache uses a direct mapping with a block size of the four words.
   a) How many bits are there in TAG, BLOCK and WORD fields?
   b) How many blocks can cache can hold? [8+7]

4. Draw and explain the 8086 CPU pin diagram. [15]

5.a) How to evaluate arithmetic expressions? Explain with examples.
   b) Convert 16-bit HEX number to equivalent BCD number using 8086 instruction set. [7+8]

6. What are the basic differences between a branch instruction, a call-subroutine instruction, and program interrupt? [15]

7. A DMA module is transferring the characters to memory using cycle stealing, from a device transmitting at 9600 bps. The processor is fetching instructions at the rate of 1MIPS. By how much will the processor be slowed down due to DMA activity? [15]

8. With the help of a neat diagram explain the memory hierarchy. [15]

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